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RoamFab: A Design Tool for Reconfiguring Parameterized Mechanisms to 3D Models With Structural Optimization

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ABSTRACT

Creating personalized 3D printing objects has become increasingly popular with the advance in end-user modeling tools and fabrication techniques. However, it remains challenging for novice makers to design and fabricate functional objects with mechanical motion such as linear or rotational. To empower users to add mechanical movements to their models for particular needs, we first investigate the commonly used 3D printable mechanisms and parameterize four mechanical primitives. We then provide a computational approach to reconfigure the mechanical primitives to user-provided 3D models, with an underlying structural optimization procedure according to a specified bending force. We further showcase a set of design examples and conduct a user study to demonstrate the potential of our approach in creating personalized functional artifacts.

1. Introduction

The recent advancements of 3D modeling tools and consumer-grade 3D printers have made personal fabrication more and more accessible, enabling end-users to design and fabricate a wide range of functional objects, e.g., connectors (Koyama et al., 2015), enclosures (Ramakers et al., 2016), adaptations (Chen et al., 2016; Sun et al., 2021), actuating mechanisms (Li et al., 2019, 2020) and folded artifacts (Noma et al., 2020; Yu et al., 2017). However, creating objects with mechanical movements remains challenging and laborious for non-experts with limited knowledge of 3D modeling and mechanisms (Berman et al., 2020). Specifically, a typical difficulty is to choose and design appropriate mechanisms for certain degrees of freedom while considering fitting dimensions and 3D printing tolerances (Calì et al., 2012; Hudson et al., 2016). Meanwhile, the final prototype inevitably encounters structural weakness after printing (Stava et al., 2012; Umetani & Schmidt, 2013; Xie et al., 2015), especially for load-bearing usage scenarios.

Prior research focused on generating joints to connect parts in models from scratch (Koo et al., 2014), extracting and recombining mechanical parts (Roumen et al., 2018), and retargeting specific mechanisms to existing 3D models (Calì et al., 2012; Zhang et al., 2017), without taking into account the commonly used 3D printing mechanisms in the hobbyist communities, as well as adapting them in user-customized ways for desired functional preferences and structural strength.

To address these challenges, we first investigated the most frequently used mechanical parts in 3D printed artifacts from online communities such as Thingiverse (2022). We then present four parameterized mechanical primitives consisting of *hinge mechanisms*, *spherical joint*, *guide mechanisms*, and *ratchet mechanisms* for linear or rotational movements. Next, we introduced a geometry generation pipeline that reconfigures our mechanical primitives into arbitrary 3D models for desired functionality. To avoid novices creating structurally fragile mechanisms which could easily break, we experimented with measuring the maximum bending stress of the critical cross-section by varying printing parameters such as infill pattern and density. In light of the experimental results, we can calculate the maximum bending strength of a mechanical part. We also provide an optimization procedure to improve its bending strength according to user-specified force while maintaining minimum filament consumption.

In this paper, we present *RoamFab*, a design tool implemented within Rhino (McNeel, 2010) that allows novices and designers to rapidly convert personalized 3D shapes into fabrication-ready models with desired mechanical functionality, with an underlying structural optimization procedure. After selecting a mechanical primitive according to the desired degree of freedom, users can simply assign a point or a surface as input for generating the mechanism geometry. They can also adjust the design parameters of the mechanism for particular needs and aesthetics. Before 3D printing, users are informed with pre-knowledge about the maximum bending strength that a mechanical part can

provide, which they can utilize to estimate how much weight the part can hold. To demonstrate the practicability of our approach, we showcase a series of artifacts and conduct a user study with our tool.

This study makes the following contributions in the area of personal fabrication and computational design: (i) creating four parameterized mechanical primitives frequently used for 3D printing objects to support different mechanical movements; (ii) proposing a structural optimization method to improve the bending strength of a 3D printing object while reducing filament consumption; (iii) presenting a geometry generation pipeline that allows end-users to reconfigure our mechanical primitives to existing 3D models for desired functionality; and (iv) validating our approach and expanding its design space in daily usage scenarios through design examples and a user study.

2. Related work

We build upon prior work in parameterized 3D printing mechanisms, computational design of functional mechanical objects, and structural analysis for 3D printing objects, which we discuss below.

2.1. Parameterized 3D printing mechanisms

Mechanisms are configured by parameters to ensure physically valid designs while maintaining geometric constraints. With commercial 3D printers, researchers have explored converting static artifacts into dynamic objects with various parameterized mechanisms, including joints (Bächer et al., 2012; Cali et al., 2012), gear trains (Zhang et al., 2017), linkages (Megaro et al., 2017; Zhang et al., 2021), telescoping structures (Yu et al., 2017), and springs (He et al., 2019; Ion et al., 2017). For example, Cali et al. (2012) derived various non-assembly joint designs such as revolute joints and ball joints. Koyama et al. (2015) used parameters to optimize the shape and grip strength of mechanical holders. Although Thingiverse (2022) has a built-in web app called *Customizer* for adjusting parametric designs, it has not established a geometric relationship with users' personalized models. RoamFab aims to integrate parameterized mechanisms to given 3D models for desired mechanical functionality in user-customized ways.

2.2. Computational Design of functional mechanical objects

In order to allow non-experts users to create functional objects, an increasing amount of research has focused on reducing the requirements for 3D modeling skills and mechanical knowledge. The vision of these studies could be generally classified into two categories: retrofitting existing objects in the physical world and functionalizing user-specified 3D models.

The first perspective aims to augment everyday objects to assist people with daily living tasks. For example, AutoConnect (Koyama et al., 2015) generates connectors for

attaching two arbitrary objects in a desired configuration. Reprise (Chen et al., 2016) and ShrinCage (Sun et al., 2021) generate useful adaptations to objects according to user-specified actions. Facade (Guo et al., 2017) introduces a crowdsourced fabrication pipeline to assist visually-impaired people in making tactile buttons. Robiot (Li et al., 2019) allows users to fabricate add-on mechanisms to actuate legacy static objects.

From the second perspective, research follows an interactive modeling approach to functionalize given 3D models by inserting assembly-free joints for articulated models (Bächer et al., 2012; Cali et al., 2012; Ureta et al., 2016), creating transformables for skeleton-based models (Huang et al., 2016; Yuan et al., 2018) and generating robotic structures (Li et al., 2020; Megaro et al., 2015). As for context-based authoring tools, Koo et al. (2014) allowed users to specify functional relationships between parts to produce a working model from scratch. Graftor (Roumen et al., 2018) helps users to extract mechanisms of rotating axles from 3D printable machines and recombine them into the target machine. Romeo (Li et al., 2020) generates a robotic arm inside a 3D model to perform user-defined tasks. Ondulé (He et al., 2019) creates parameterizable deformation behaviors in models with helical springs and embedded joints. The above methods focus on specific mechanisms to augment the default functionality.

RoamFab extends prior work by empowering end-users to configure a wide range of mechanical movements to personalized static 3D models. We seek to help novices without expertise in mechanism overcome the barriers between mechanical design to desired functionality, thereby contributing domain-specific knowledge of 3D printable mechanisms and a novel design workflow in creating such functional artifacts.

2.3. Structural analysis for 3D printing objects

Traditionally, the Finite Element Method (FEM) is conducted to simulate the bending behavior of a given 3D model when optimizing its shape for desired behavior (Chen et al., 2015; Xie et al., 2015; Xu et al., 2015). However, FEM-based shape modeling works in a trial and error way, thus requiring professional knowledge from novice makers (Zhang et al., 2016). Therefore, we seek to directly inform users of the maximum bending strength of a mechanical part with existing parameters and strengthen it according to user-specified load at interactive rates.

In recent years, many computational design methods have been developed to provide users with interactive controls over the structural robustness of 3D printing objects, including shape deformations (Stava et al., 2012; Zhou et al., 2016), modifications to the interior structure (Lu et al., 2014; Stava et al., 2012), and changing of printing directions (Hildebrand et al., 2013; Umetani & Schmidt, 2013). Stava et al. (2012) improved the structural strength of objects through hollowing, thickening, and strut insertion. Lu et al. (2014) introduced an optimization algorithm for re-shaping the interior structures of a model to relieve the internal stress. Umetani and Schmidt (2013) proposed a cross-sectional structural

analysis technique that efficiently detects critical stress inside a 3D object based on bending momentum equilibrium. They optimized the orientation for 3D printing to increase mechanical strength.

Our structural optimization procedure also focuses on the critical cross-sections of beam-like mechanical parts by optimizing the 3D printing and shape parameters. We conducted a mechanical experiment to study how different printing parameters influence the maximum bending stress. The obtained data is used to predict and optimize the bending strength of generated mechanisms. A similar data-driven approach has been used to determine the relationships between bending elasticity, shell thickness, and cross-section dimensions (Zhang et al., 2016).

3. Mechanical primitives

To reduce the gulf of execution (Norman, 2013) that non-experts must bridge from designing mechanisms to desired mechanical movements such as linear or rotational, we present a set of parameterized mechanical primitives, including hinge mechanisms, spherical joint, guide mechanisms, as well as ratchet mechanisms. This section describes these mechanisms and their design rationale. Detailed usage scenarios of them are discussed in Section “Design Examples.”

3.1. Design goals

To ensure that artifacts created by RoamFab can be utilized in most daily life scenarios, e.g., gadgets, hand tools, containers, or toys, we derived our mechanical primitives by searching for the most frequently used mechanisms in makers communities, such as Thingiverse (2022), Pinterest (2022), and Prusaprinters (2022). We used “most downloaded” and “most makes” to sort the results. Additionally, the design parameters of each mechanical primitive can be customized for motion requirements, such as the angle of rotation or the range of movement. Shape parameters can also be optimized to fit the dimensions of user-specified 3D models while preserving mechanical functionality. Additionally, all mechanical primitives are printed with default parameters and tolerances to test their working, feasibility to print-in-place, and easy assembly.

3.2. Hinge mechanisms

To enable two objects to rotate relative to each other, RoamFab provides three types of hinge mechanisms shown in Figure 1. The first hinge mechanism embeds two separated parts for revolute motion and can be 3D printed in place along the rotating axis. For conditions where the user needs to assemble a hinge manually, we provide the second hinge mechanism consisting of three parts, where the ISO metric screw thread is used to hold the two parts together. The third hinge mechanism, also known as a cylindrical joint with two degrees of freedom, can be used when the user does not need the limit parts at both ends of the rotating shaft to provide a single-axis sliding function. Essential

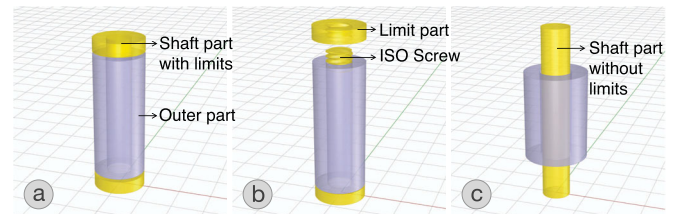


Figure 1. The three types of hinge mechanisms with 6, 7, and 5 design parameters respectively.

design for hinge mechanisms is the printing tolerance between the inner part and the outer part that allows them to hinge.

3.3. Spherical joint

RoamFab offers a spherical joint design consisting of a ball and socket, which allows rotational movements with three degrees of freedom, as shown in Figure 2. The two openings in the socket enable the ball to be pressed into it by elastic (Figure 2(a)). Figure 2(b) shows the solid part that can be subtracted by an arbitrary shape to create a socket. We tested five different tolerances from 0.03 mm to 0.15 mm between the ball and socket to provide internal friction for supporting static poses under any rotation through five sets of trials, eventually settling for 0.06 mm as default. We did not choose an assembly-free design of spherical joint used in (Cali et al., 2012) because whether the connected objects can be printed together is unpredictable.

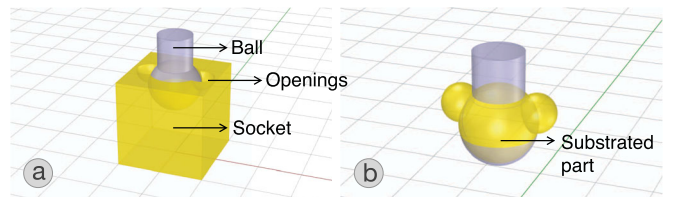


Figure 2. The spherical joint with 2 design parameters.

3.4. Guide mechanisms

RoamFab supports two types of guide mechanisms for translational movements between objects along a straight or curved path, shown in Figure 3. The first guide mechanism is a prismatic joint that allows linear motion, constraining revolute motion by the polygonal cross-section. For many applications of guide mechanisms, it is desirable for objects to move along a user-customized curved path. Therefore, we designed the second guide mechanism by replacing the polygonal-shaped slider with a ball, thus providing the ability to translate along a curved rail and rotate about an axis normal to the plane, bringing two degrees of freedom. Additionally, the limiter at both ends of the guide rail prevents the slider from moving out of the guide rail, which is determined by the user to use or not. If the limiter is added, it would be joined together with the guide rail as one solid geometry.

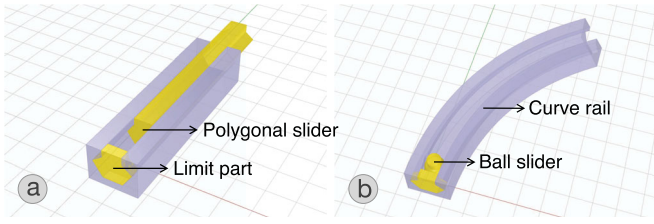


Figure 3. The two types of guide mechanisms with 4 and 5 design parameters respectively.

3.5. Ratchet mechanisms

Ratchet mechanisms allow rotary motion in one direction with discrete steps and prevent movement in the reverse direction, which can be used to handle loads. RoamFab offers a ratchet mechanism with an inner pawl that is squeezed when rotating by exploiting the elasticity of PLA (Figure 4). Since the pawls are the most fragile structures, we conducted a series of tests with different thickness pawls and set the minimum thickness to 1.2 mm. This size ensures that the inner pawl can be smoothly rotated and will not break without significant destructive force in the opposite direction. To increase the load the ratchet can handle in the reverse direction, the user can increase the depth of the sawtooth or the height of the ratchet wheel. The sawtooth count on the ratchet wheel can be customized for different degrees of the swing arc. As shown in Figure 4(b), we extend this primitive with a cylindrical shape for aesthetic reasons as generated in the design example of the solder spool holder (Figure 16). At the other end, symmetrical to the ratchet, we use a screw nut to prevent the pawl from getting out of the ratchet wheel.

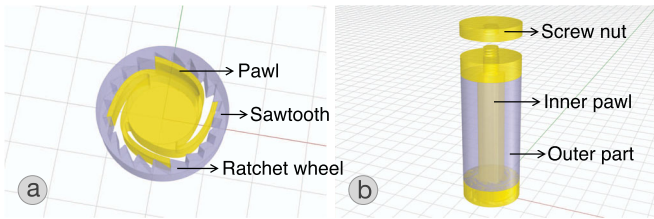


Figure 4. The ratchet mechanisms with 6 design parameters.

4. Structural optimization

Functional 3D printing artifacts often have strength requirements such as supporting or holding everyday objects or being used as hand tools. We aim to inform novice makers of the allowable bending strength of the mechanical primitives while configuring them, thus avoiding creating fragile objects. Informed by the cross-sectional structural analysis (Umetani & Schmidt, 2013) based on the Euler-Bernoulli assumption, which is widely used in designing structures under bending deformation, we further conducted a mechanical experiment to analyze the relationships between the maximum bending stress inside a cross-section and different 3D printing parameters. After the data acquisition of the maximum bending stress of a beam-like primitive shape, we provided a data-driven optimization procedure and integrated it into RoamFab, i.e., when fabricated by the same material used in our experiment, the model's bending strength can be predicted with pre-known printing parameters and cross-section dimensions.

4.1. Mechanical experiment

Objects manufactured via layer-by-layer methods such as the Fused Deposition Method (FDM) exhibit significant structural anisotropy. Although objects can bear considerable tensile forces along the direction of filament accumulation, they are especially fragile when the applied force is perpendicular to the orientation (Es-Said et al., 2000). When the beam is bent and deformed, fracture occurs if the external force produces stress that exceeds the maximum bending stress that the critical cross-section can bear. The maximum bending stress depends on the material properties such as Young's modulus and printing parameters (Alvarez et al., 2016; Khan et al., 2018). Our experiment aims to study the maximum bending stress inside the cross-section when the force is perpendicular to the filament orientation by varying infill patterns and densities. For all tests, we break cantilever beams printed with PLA using a universal testing machine, shown in Figure 5(a). We calculate the maximum bending stress σ_{max} by the following equation based on moment equilibrium (Budynas & Nisbett, 2011):

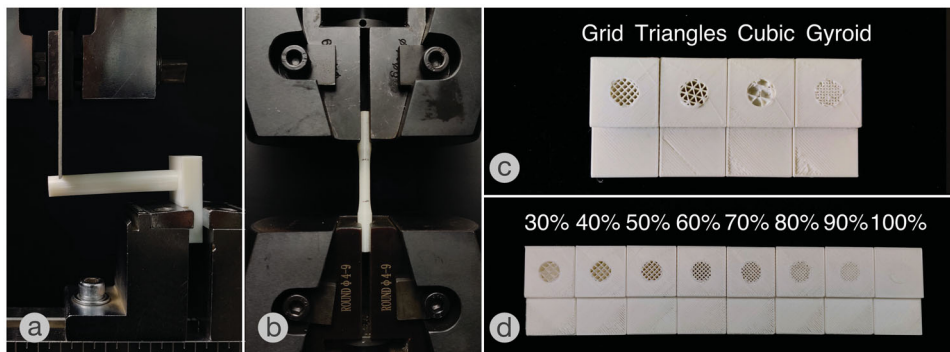


Figure 5. Mechanical experiment. (a) The cantilever beam used in the bending strength test. The arm length is 57.84 mm, and the radius of cross-sections on the beam is 5 mm. Other printing parameters include 0.2 mm layer height and 0.4 mm wall thickness. (b) The test specimen used in the tensile strength test. (c) The four different infill patterns with 50% infill density. (d) The different infill densities of the grid infill pattern.

Table 1. Results of infill pattern test.

Infill density (%)	Measurements	Grid	Triangles	Cubic	Gyroid
30	Force (N)	15.63	14.80	13.22	11.35
	σ_{max} (Mpa)	9.21	8.72	7.79	6.69
50	Force (N)	24.30	23.41	19.43	16.53
	σ_{max} (Mpa)	14.32	13.79	11.45	9.74
70	Force (N)	29.07	28.70	25.64	23.94
	σ_{max} (Mpa)	17.13	16.91	15.11	14.10

$$\sigma_{max} = \frac{M}{I} \times y_{max} \quad (1)$$

where the bending moment $M = FL$, F is the external force, L is the arm of force, y_{max} is the farthest distance between points on the cross-section to the neutral axis, and I is the area moment of inertia that defines the shape-based resistance of a cross-section to bending. Since the fractured cross-section in our test is circular with area A and radius R , we can get I by:

$$I = \int_A y^2 dA = \frac{\pi R^4}{4} \quad (2)$$

4.1.1. Infill pattern test

We compared four different infill patterns, including grid, triangles, cubic, and gyroid, each of which has 30%, 50%, and 70% infill densities (Figure 5(c)). The first two infill patterns are known for providing strength in two dimensions, and the latter two can provide strength in three dimensions (O'Connell, 2022; PrusaSlicer, 2022). Results are showed in Table 1. Among them the grid infill performs the best, which can be explained by this pattern having the largest contact area between layers, thus providing better layer adhesion. We conclude that although the same infill density guarantees the consistent consumption of filament, the infill pattern still contributes to bending strength.

4.1.2. Infill density test

Since the grid infill outperforms other infill patterns while bending, we further tested its bending strength across different infill densities from 30% to 100%, shown in Figure 5(d). We used the average of three sets of samples to get the final results. Table 2 shows that the maximum bending stress increases as infill density increase with more filament within a layer.

Additionally, it is worth noting that we also tested the tensile strength under which layers of filament separate (Figure 5(b)), according to ASTM standards (2015). Even if the infill density is set to 30%, the test specimen can withstand a force of 85.33N, suggesting that it is hard to break 3D printing objects with tensile forces. Therefore, we conclude that our structural optimization procedure focuses on improving the bending strength of our mechanical primitives in the reconfiguration process.

4.2. Optimization procedure

Based on the static strength test results for maximum bending stress, we use the grid infill pattern and 30% infill density as the

Table 2. Results of infill density test.

Measurements	30%	40%	50%	60%	70%	80%	90%	100%
Force (N)	15.75	19.58	24.41	26.84	29.05	31.68	34.50	44.46
σ_{max} (Mpa)	9.28	11.54	14.38	15.81	17.11	18.66	20.33	26.19

default 3D printing parameters in RoamFab. Since we already know the cross-section dimensions of a mechanical part, we can predict its maximum bending strength according to Equation 3, as long as the PLA material is used for fabrication.

Considering the trade-off between mechanical strength and filament consumption, RoamFab aims to improve the bending strength of the generated mechanical part to satisfy a user-specified bending force while consuming the least materials. Given an applied bending force F and the length of the arm L , we first find the required allowable bending stress $[\sigma]_r$ by:

$$\frac{M}{L} = \frac{\int_A y_{max} [\sigma]_r dA}{L} \geq F \quad (3)$$

Then we optimize the 3D printing and shape parameters of the generated mechanical part to ensure the critical cross-section can provide the required allowable bending stress $[\sigma]_r$. We start with optimizing the default infill density of the cross-section at the end of the beam with moment arm L , to increase its maximum bending stress, i.e., we can use $[\sigma]_r$ to obtain the specific infill density R from Table 2. In our calculation, we use a default safety factor $s = 1.25$ to get the allowable bending stress $[\sigma]$ of each infill density by: $[\sigma] = \sigma_{max}/s$. If the cross-section with 100% infill still cannot meet the force, we then increase the area of the cross-section according to Equation 2 by optimizing the shape parameters of the mechanical part. These two steps guarantee that we can satisfy the required allowable bending stress.

Next, we use a gradient approach to optimize the infill density in the 3D printing process to reduce filament consumption, as the force on the cross-section of the beam decreases as the arm of force becomes smaller. Specifically, we gradually reduce the infill density R by 10% along the printing direction until it reaches 30% infill (Figure 6). To ensure that fracture does not occur in the middle of the beam, we calculate the printing height ΔL_1 before changing to a smaller infill density with allowable bending stress $[\sigma]_{r-1}$ by:

$$\Delta L_1 = L - L \frac{[\sigma]_{r-1}}{[\sigma]_r} \quad (4)$$

We also observe that the contact area is very small between layers where infill density changes. Therefore, we use 100% infill on the top and bottom layers of each infill

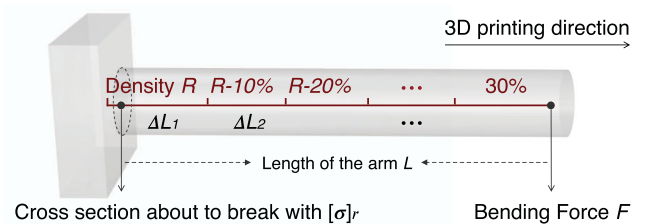


Figure 6. Gradually reduce the infill density along the printing direction for minimum filament consumption.

block. We further validate our gradient approach using cantilever beams with infill densities from 40% to 100%. These specimens have roughly the same bending strength as before optimizing the filament consumption, and no fracture occurs in the middle of beams. The result demonstrates that our optimization procedure can improve the bending strength of beam-like mechanical parts and minimize the use of filament.

5. Reconfiguration process

RoamFab aims to empower end-users to reconfigure our parameterized mechanical primitives to fit arbitrary 3D models for desired functionality through intuitive interactions. The final output is a working prototype that can be fabricated with consumer-grade 3D printers and is easy to assemble. As we discuss below, designing with RoamFab is an iterative process and involves five steps: (1) specifying input, (2) generating the initial design, (3) customizing design parameters, (4) optimizing structural strength, and (5) generating fabrication-ready 3D models. Once satisfied, the user can export all the geometry as 3D printable STL files. Otherwise, users can still modify their models for regeneration.

5.1. Specifying input

Users start with importing a 3D model that should be a closed polysurface in Rhino, and the system will duplicate the model to avoid direct modification. After users select a mechanical primitive based on the desired degree of freedom of mechanical movement (Figure 7(a)), a new parametric instance is initialized with default parameters. Next, RoamFab interactively guides users to specify where and how the mechanical primitive should be configured into the model (Figure 7(b)).

At first, users are asked to specify a pivot point on the model (Figure 7(c)) to help the system get the coordinates to place the mechanisms (Figure 7(f)). The pivot point is

regarded as the lowest point to place the generated geometry. However, the pivot point becomes not necessary if users select guide mechanisms, which instead require them to specify a line or curve as input to generate the guide rail (Figure 7(g)). Besides, a unit vector along X/Y/Z axes is an essential input used as the orientation to generate mechanisms. Sometimes users need to split the model into two parts for respectively combining mechanisms such as hinge and spherical joint for relative movements. They also need to create a cutter surface (Figure 7(d)), by which the system will automatically split the target model and caps planar holes, resulting in two closed polysurfaces (Figure 7(e)). Note that the pivot point should also lie on the cutter surface.

5.2. Generating the initial design

According to user input, RoamFab generates the initial design of mechanisms by performing a geometric analysis of the user-provided 3D model for constraining parameters. This process starts with extracting a reference surface and two reference vectors, denoted as V_L and V_S . The reference surface is extracted from the target model while ensuring that the pivot point lies on it and the normal vector of the surface is parallel to the user-selected unit vector (Figure 8(a)). If the reference surface is not found on the model, the system automatically creates it by finding the intersection of a plane and the target model. To obtain the two reference vectors, the system creates a line passing through the pivot point and parallel to the unit vector, and then uses the intersection points with the model to compute the two reference vectors that have the largest (V_L) and shortest (V_S) lengths (Figure 10(b)). Below we describe how to utilize the reference surface and the reference vectors to calculate the design parameters of different mechanisms.

5.2.1. Hinge mechanisms

While generating hinge mechanisms, the cross-sectional area of the hinge is constrained by the area of the reference

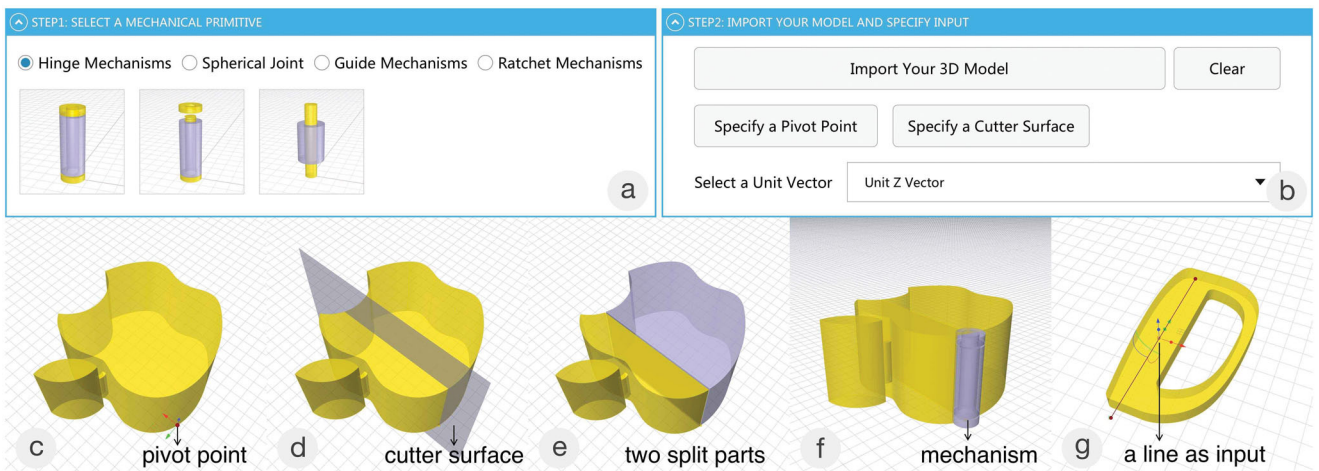


Figure 7. (a) The user interface for selecting a mechanical primitive. (b) The user interface for importing a 3D model and specifying input. (c) Specify a pivot point on the target model. (d) Specify a cutter surface to split the model. Ensure that the pivot point lies on the cutter surface. (e) The system splits the model into two parts based on the cutter surface. (f) The mechanism is placed on the pivot point. (g) A line is required for generating guide mechanisms.

surface with a proportional relationship (Figure 8(a)). The length of V_L serves as the hinge length (Figure 7(d)). For models that are split to combine hinge parts, the system moves the two pieces a millimeter away from each other to avoid adhesion while printing (Figure 7(c)). In order to create space for the generated mechanism, a boolean difference operation is performed by the target model and a solid cylinder generated by the system (Figure 8(b)). When the pivot point locates on the edges of a model (Figure 8(c)), RoamFab allows users to choose whether to put the hinge mechanism tangent to the edges of the target model (Figure 8(d)). Based on the pivot point and the radius of the hinge, the system computes a new pivot point to place the generated geometry.

5.2.2. Spherical joint

The spherical joint is placed on the pivot point by default (Figure 9(b)). Moreover, embedding spherical joints often involves users in specifying a cutter surface to split the model, then the system would extract the centroid point on the intersection surface to place the mechanism (Figure 9(d)). The radius of the ball is proportional to the area of the reference surface with a 3 mm minimum threshold to maintain functionality. The length of V_S is also used to ensure the generated mechanism is inside the target model (Figure 9(b)). RoamFab supports generating multiple spherical joints on a model simultaneously by utilizing a set of cutter surfaces (Figure 9(c)). The orientation of the generated spherical joints is the normal vector of the specified cutter surfaces. Users can also reverse the position of the ball and socket for connecting with different parts (Figure 9(e)).

5.2.3. Guide mechanisms

To embed guide mechanisms, a user-specific straight line or a curve is used as a geometry parameter (Figure 7(d)), along which the system sweeps a guide rail with a polygonal-shaped slider or ball slider. The position of the line or curve is required to be on or inside the target model. Then the length of V_S is used to constrain the height of the guide rail. Same as using spherical joints, users can also reverse the orientation of the generated mechanism, then the part of the target model connecting with the slider is movable.

5.2.4. Ratchet mechanisms

The radius of the ratchet wheel is also proportional to the area of the reference surface (Figure 10(a)), with a minimum threshold of 12 mm to guarantee its working. Then the thickness of the ratchet wheel is obtained by the length of V_S . The height of the generated mechanical part is equal to the length of V_L minus the thickness at both ends (Figure 10(c)). Same as configuring hinge mechanisms, a solid cylinder is subtracted by the target model to create space for the generated ratchet mechanism (Figure 10(b&e)), and the placement of the mechanism is default to be tangent to the edges on the model where the pivot point is located (Figure 10(f)). Users can reverse the orientation of the generated mechanism for clockwise or counterclockwise rotation.

5.3. Customizing design parameters

With the initial generation of mechanisms, users can first drag the pivot point or the cutter surface in Rhino to interactively change the generated mechanism to a new desired position (Figure 11(a&b)). Then users can further fine-tune the design parameters of the generated mechanisms for

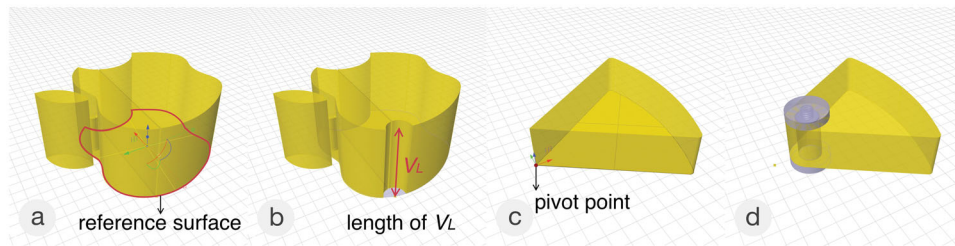


Figure 8. (a) The extracted reference surface is used to constrain the cross-sectional area of the hinge mechanism. (b) The length of the largest reference vector V_L . A solid cylinder is subtracted by the target model to create space. (c) The pivot point locates on the edges of the target model. (d) Put the generated hinge mechanism tangent to the edges.

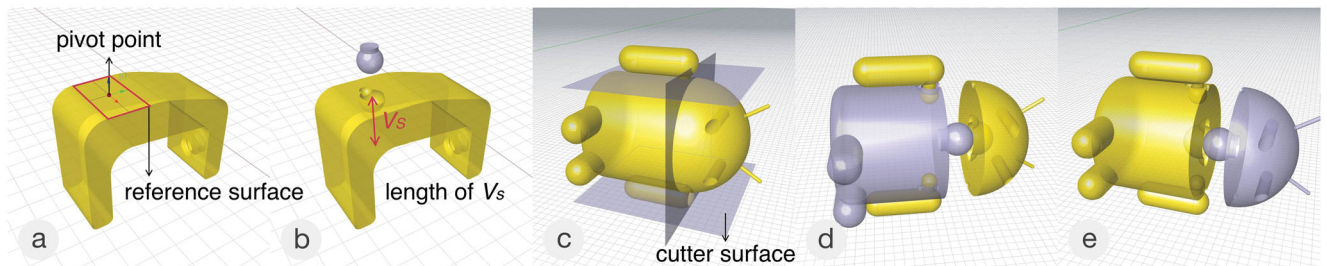


Figure 9. (a) The input pivot point and the extracted reference based on it. (b) The length of the smallest reference vector V_S . (c) Specify multiple cutter surfaces. (d) The system uses the the centroid point to place the spherical joint. (e) Users can reverse the relative position of the ball and socket.

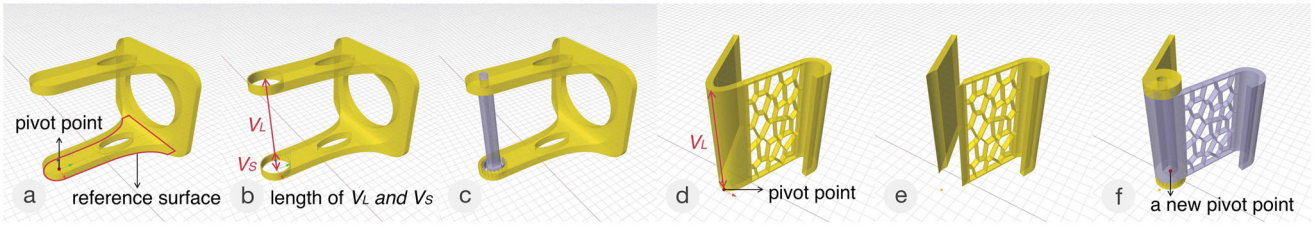


Figure 10. (a) The input pivot point and the extracted reference based on it. (b) The length of the largest reference vector V_L and the smallest reference vector V_S . (c) Generating the mechanical part based on V_L and V_S . (d) The pivot point on the target model. (e) The target model is divided into two parts after subtracting a solid cylinder. (f) Put the generated ratchet mechanism tangent to the edges.

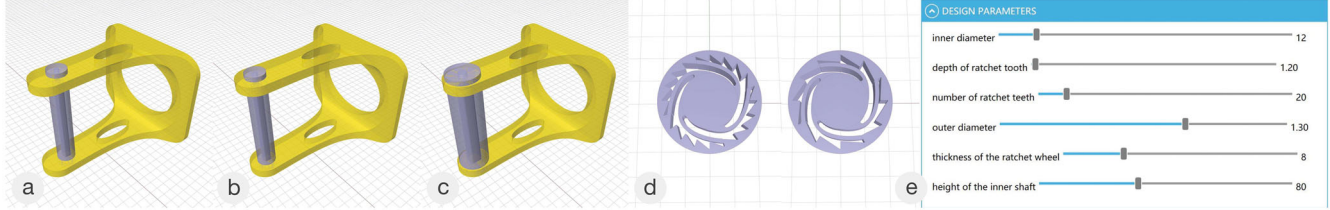


Figure 11. (a) An initial position. (b) Move the generated mechanism to a new position by dragging the pivot point. (c) Increase the radius of the stressed part to make it stronger. (d) Adjust the number of ratchet tooth. (e) Slider widgets for adjusting the design parameters of the ratchet mechanism.

individual needs, e.g., increasing the radius of the stressed part to make it stronger (Figure 11(b)), adjusting the number of ratchet teeth for different degree of the swing arc (Figure 11(d)), or adapting the width of a guide rail to an object (Figure 15(a)). This can be done via slider widgets or entering specific values. Explicit explanation of design parameters is shown in the user interface (Figure 11(e)). All changes are reflected in real-time in the Rhino 3D modeling environment.

5.4. Optimizing bending strength

RoamFab supports two stress conditions of our mechanical parts. Specifically, the applied force is at the end or in the middle of the beam, as shown in Figure 12. When the user adjusts the design parameters of the generated mechanisms, RoamFab provides a real-time calculation of the maximum bending strength with the default grid infill pattern and 30% infill density according to Equation 3. Therefore, the user can estimate the weight that the mechanical part can bear. RoamFab also supports the user to specify the magnitude of the force F applied on the mechanical part. Then the system follows our optimization procedure to satisfy the required bending strength, i.e., optimizing the printing and shape parameters. Note that when the cross-section at both ends of the beam is under force, the gradual optimization of infill density is used at both ends. RoamFab also informs users of the maximum bending moment that cross-sections on the

beam can provide, which they can use to calculate the bending strength of a mechanical part according to the specific arm length L .

5.5. Generating fabrication-ready 3D models

Once the user has completed the customization, they can click a button to inform the system. Then RoamFab will conduct several post-processing steps to generate fabrication-ready 3D models, such as creating printing tolerances to avoid junction (Figure 13(a)), scaling part of the target model to fit the generated mechanical part (Figure 10(e)), or performing boolean operations between mechanical parts and the target model. For a model that is divided into two parts, RoamFab combines each of them with different mechanical parts for relative movements. We obtain the relative positions of different objects by calculating their volume centroids to ensure correct boolean operations. However, the system would not generate fabrication-ready designs if the original 3D model does not have enough volume to accommodate the generated mechanisms at their minimum parameter thresholds.

6. Implementation and fabrication

We create our parameterized mechanical primitives using Grasshopper (2022), a visual programming language and environment that runs within Rhino. Then we encapsulate

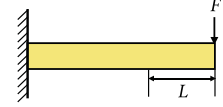
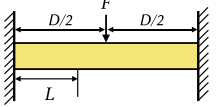
Stress Condition	Bending Moment (M)	Stress Condition	Bending Moment (M)
	$M = FL$		$0 \leq L \leq \frac{D}{2} \rightarrow M = \frac{FD}{2} \left(\frac{1}{4} - \frac{L}{D} \right)$ $L = 0; L = \frac{D}{2} \rightarrow M = \pm \frac{FD}{8}$ $\frac{D}{2} \leq L \leq D \rightarrow M = \frac{FD}{2} \left(\frac{L}{D} - \frac{3}{4} \right)$

Figure 12. The first stress condition is the applied force at the end of the beam. The second stress condition is the applied force in the middle of the beam.

each mechanism as a cluster and export it as a.ghcluster file. Therefore, we can initialize them as the Grasshopper Data Tree with default parameters. The imported 3D model and generated mechanisms used in our design examples and the user study are boundary representation (BREP) objects. If the user integrates a mechanism to a mesh geometry, the BREP mechanism will be automatically converted to mesh for boolean operations. We implement all the geometry processing functions using RhinoPython and RhinoCommon API. We also use Human UI (Andheum, 2022), a plugin for Grasshopper, to develop the user interface. Everything runs on an Alienware computer with a Windows system. All of the artifacts created in our work are 3D printed by Creality Sermoon D1 using PLA.

To print the mechanical part that uses our gradual optimization of infill density to improve its bending strength, we use Ultimaker Cura 4.12 (2022) to “modify printing settings for overlaps” between the part and several cubes. The number of cubes depends on the number of times the infill density changes. We set the height of each cube according to its corresponding value ΔL_n . A top and bottom layer are set for each cube to avoid too small contact area when infill density changes. Other printing parameters are consistent with the non-overlapping parts of the mechanical part.

7. Design examples

To highlight an initial design space and demonstrate the breadth of functional artifacts created by RoamFab, we present nine examples using our mechanical primitives and reconfiguration pipeline.

7.1. Hinge mechanisms based examples

We showcase three functional examples in Figure 13 that embed the three types of hinge mechanisms we described in Figure 1. To create the apple-shaped box in Figure 13(a) with the top of it can flip backward, a pivot point and a

cutter surface are required. The generated mechanical parts are joined respectively with the split two pieces of the original shape. Tolerances are also generated to guarantee its functionality while printing in place (Figure 13(a2)). The second example is a pillbox with a hinge that needs to assemble. When the screw nut is tightened, the lid of the box is fixed. For aesthetic considerations, RoamFab lets users choose whether to configure the mechanism to be tangent to the edges of the target model (Figure 13(b2)). The laptop storage rack in Figure 13(c) is an example that embeds the cylindrical joint with no limits at both ends. Since the inner shaft of this mechanism needs to bear the weight of the placed laptop, we use our structural optimization procedure to ensure the critical cross-section can provide enough bending strength. Note that this example can also be printed in place.

7.2. Spherical joint based examples

Figure 14 shows two examples with spherical joints. The first example clips a flashlight for flexible adjustments of its angle. With our pre-set tolerances, the socket can handle any static poses of the flashlight while rotating. With RoamFab, we are informed in advance that the cross-section at the end of the beam can withstand the weight of the flashlight with a default 30% infill density before printing. A pivot point on the clip is required to place the mechanism (Figure 14(a2)), and the connector between the ball and the flashlight is added after reconfiguration. The second example is an Android robot embedding three spherical joints to support movements of its head and arms, which requires three cutter surfaces as input.

7.3. Guide mechanisms based examples

Two examples are showed in Figure 15 based on our two types of guide mechanisms (Figure 3) for linear movements. The blade handle hides the blade inside and slides it out

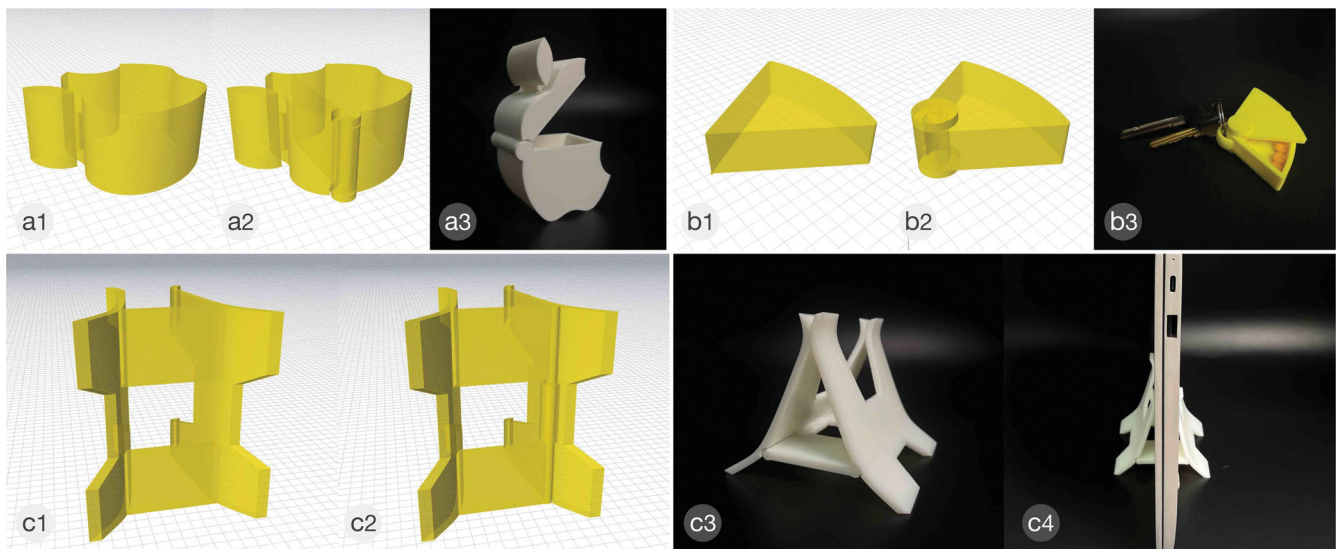


Figure 13. The figure shows the design examples with their initial shapes and fabrication-ready models that embed hinge mechanisms. (a) An apple-shaped box. (b) A pillbox. (c) A laptop storage rack.

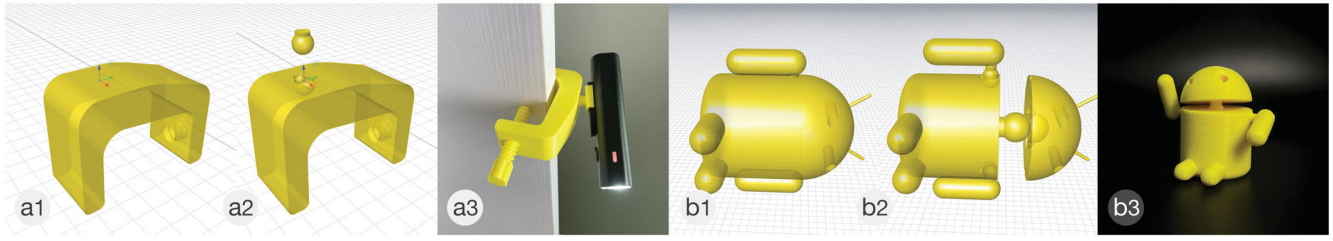


Figure 14. The figure shows the design examples with their initial shapes and fabrication-ready models that embed spherical joints. (a) A flashlight clip. (b) An Android robot.

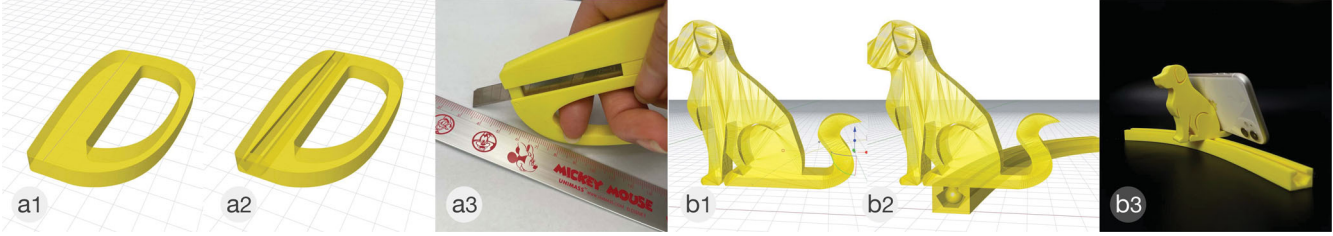


Figure 15. The figure shows the design examples with their initial shapes and fabrication-ready models that embed guide mechanisms. (a) A blade handle. (b) A camera slider.

when in use. RoamFab performs boolean operations to create the guide rail inside the handle. The slider is used for pushing or retracting the blade. The limit part on the rail can prevent the blade from falling out of the handle. We fine-tune the design parameters of the mechanism to fit the dimensions of the blade. We also demonstrate a camera slider with a curved rail for shooting videos. The ball slider is automatically joined with the dog-shaped phone holder.

7.4. Ratchet mechanisms based examples

As shown in Figure 16, we demonstrate two examples using ratchet mechanisms for rotational motion in only one direction. The first example is a stand using which the tilt angle of a Kindle Paperwhite can be adjusted while reading (Figure 16(a3)). The ratchet handles the weight of the book to prevent movement in the opposite direction. The part holding the book can be rotated 360 degrees for re-adjustments. A pivot point is required on the initial model, and then RoamFab automatically splits the model and scales part of it to combine different mechanical parts (Figure 16(a2)). The second example is a solder spool holder where the ratchet limits the solder wire to be dragged out in only one direction with a small discrete length (Figure 16(b3)). After specifying a pivot (Figure 16(b1)), RoamFab performs a geometric analysis of the model

to set the parameters of the generated mechanical parts, i.e., the height of the holder and the thickness of the ratchet wheel. To ensure the holder provides enough bending strength after putting the solder spool, RoamFab optimizes the cross-sections on both ends of the inner pawl part in terms of the infill density for printing and design parameters.

After three months of repeated usage of the Kindle Paperwhite stand, the inner pawl has not broken with its minimum 1.2mm thickness. However, we do observe a loss of elasticity in the structure. Overall, we assume the structural fatigue of the ratchet mechanism is not an issue in daily scenarios with our lower limit for the thickness of pawls.

8. User study

The objective of the user study is to evaluate the usability of RoamFab and help us understand its limitations for future work. We reported our qualitative observations based on participants' feedbacks.

8.1. Participants

We recruited eight participants (4 female, average age 25.5 years, $SD = 2.75$) to utilize RoamFab to create functional artifacts with desired mechanical movements, five of which

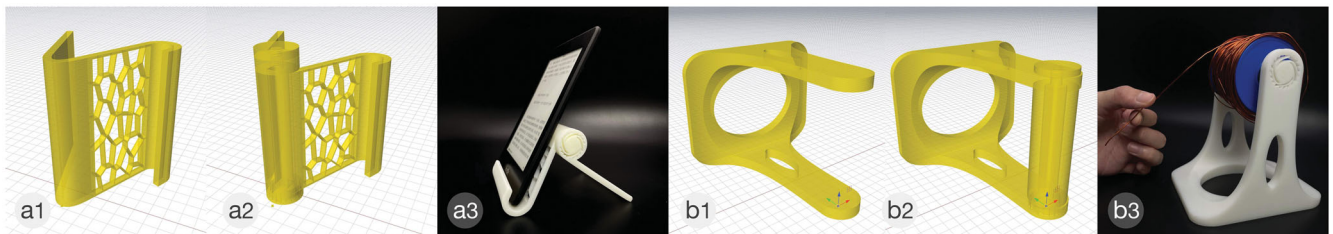


Figure 16. The figure shows the design examples with their initial shapes and fabrication-ready models that embed ratchet mechanisms. (a) A Kindle Paperwhite stand. (b) A solder spool holder.

have little 3D modeling skills using CAD tools. Three participants used Rhino before and have experience with 3D printing, among which one participant stopped using 3D printing after an initial failed attempt. Most participants have no mechanical knowledge.

8.2. Task and procedure

We assigned each participant an open-design task to utilize RoamFab to configure an existing 3D model for desired mechanical movements. The task was split into three sessions to budget time for preparing models and 3D printing.

8.2.1. Preparation session

We sent each participant an introduction file about our mechanical primitives, how they work, and the initial design space of RoamFab. Then participants were required to come up with a target model and its usage scenario where they wanted to embed one of our mechanisms. We have no restrictions on the shape of the provided models. Depending on the recruitment order, we guarantee that each of our four mechanical primitives is selected twice. At last, four participants found their target models from online communities. Three participants built their own models. One participant invited us to help with modeling after providing a sketch.

8.2.2. Design session

Participants brought their 3D models to our laboratory to perform the task. To start, participants were introduced to how to specify input to configure a model from our design

examples. Then participants practiced with the example until they claimed that they had understood the reconfiguration process of RoamFab. Next, participants conducted the formal task with their models in a think-aloud manner. We observed participants' design process and gave instructions when they were confused. After participants confirmed that they had the desired results, they attended a semi-structured interview to describe their experience, such as the difficulty of using RoamFab, satisfaction with the fabrication-ready models, encountered problems, and suggested features. We recorded the behaviors and utterances of participants throughout the design session. For models involving post-processing such as hollowing the internal volume or adding new parts for their final working, we implemented them under participants' guidance.

8.2.3. Fabrication session

We validated the functionality of created objects with a consumer-grade FDM 3D printer, and invited participants to retrieve them the next day. Then we further interviewed participants about whether the fabricated results matched their expectations.

8.3. Results

As Figure 17 shows, all participants successfully produced functional objects with specific mechanical movements. We reported our findings in terms of successes and shortcomings of RoamFab as follows.

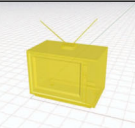
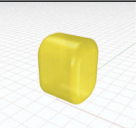
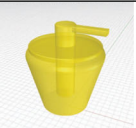
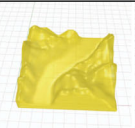
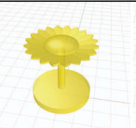
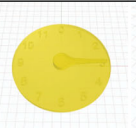
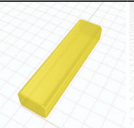
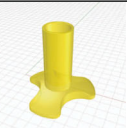
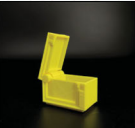
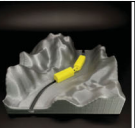
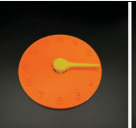
Participants	P1 (Television Box)	P2 (Earplug Box)	P3 (Stir Handle)	P4 (Miniature Landscape)	P5 (Sun-flower)	P6 (Clock Dial)	P7 (7day-Pillbox)	P8 (Pen Holder)
Levels of 3D Modeling Experience	Novice	Novice	Novice	Mid-level	Entry-level	Entry-level	Novice	Novice
Source of the Target Models	Online community	Online community	Invite us to model	Own	Own	Own	Online community	Online community
Task Completion Time (minutes)	23	35	30	26	18	29	28	24
Mechanical Primitives	Hinge Mechanism	Hinge Mechanism	Ratchet Mechanism	Guide Mechanism	Spherical Joint	Ratchet Mechanism	Guide Mechanism	Spherical Joint
Existing Models								
Reconfigured Objects								

Figure 17. Results from the user study. The relationship between different levels of 3D modeling experience is that "Mid-level" > "Entry-level" > "Novice."

8.3.1. Simplification of authoring process

Most participants highly valued that RoamFab simplifies the authoring process and reduces the expertise for creating functional objects by encapsulating parametric mechanisms and 3D printing tolerances. Thus, they only need to focus on the shape design of the 3D model and think about where and how to integrate the mechanism, without worrying about its functionality after fabrication. Also, P1 and P4 mentioned that the tool is especially useful when configuring multiple mechanisms to models. P3, P5, and P8 considered that automatically fitting parameters to geometry dimensions saves much efforts without the need to fine-tuning.

8.3.2. Reconfiguration performance

Overall, participants are positive about the initial design of generated mechanisms. P4 mentioned that *“the guide mechanism is generated exactly in the space reserved for it on the original model.”* P2, P3, and P6 especially like being able to change the position of the mechanism by dragging the pivot point. P1 was particularly fond of the feature that the system supports to place the hinge mechanism to be tangent with edges of the target model, and said, *“although the lid of the box can only be opened 90 degrees, it brings better aesthetics.”*

While customizing the mechanisms for personal preferences, participants considered that the role of different design parameters is easy to follow. P2 and P6 encountered the minimum threshold we set while reducing the size of the mechanism. Besides, P1 suggested discrete adjustments with a uniform size. P2 expected to keep the hinge mechanism centered by reducing the same dimension at both ends when adjusting its length. P4 suggested that we can highlight the color of the mechanism, making it easier to observe its shape change during the adjustment.

Participants also confirmed that the fabricated results are satisfactory and match their initial conceptions during the preparation session. They look forward to using the customized artifact in their daily life. P6 and P7 even said that the final results *“exceeded my expectations,”* because they were able to come up with two ways to embed the mechanism to their models. P5 and P8 were satisfied with the internal friction of the spherical joint.

8.3.3. Difficulty of using RoamFab

Most participants were able to grasp the design process with RoamFab and indicated their willingness to use it again to create another functional 3D printing artifact. However, we found that participants had difficulties in handling interfaces. Specifically, P2 and P3 found that it is difficult to pick a desired point on the model precisely. P8 did not know how to build a surface in Rhino. P4 and P7 stated that generating guide rails by drawing lines is intuitive. Nevertheless, P4 reminded us that drawing nice curves by control points in Rhino can be difficult for novices with little modeling experience. If they did not draw the curve in the top view but in the perspective view, the resulting curve may not be

as expected. P7 also mentioned that it is inconvenient to draw a line in the middle of a surface and align it manually.

We also observed that the major challenge is figuring out where to place the mechanism to work with the original model. As P6 stated, *“I tried and error by adjusting the position of the ratchet mechanism until I finally found the right solution.”* P1 also mentioned that *“I still need some spatial thinking to understand the relationship between the original model and the hinge mechanism, until then I just want to place it randomly, like try it out inside of the model.”* To summarize, they wish the system to recommend some surfaces or points as the pre-knowledge to configure mechanisms, from which they can choose and drag them by mouse to specify the correct input to the system.

8.3.4. Encountered problems and suggested features

Four participants expected to have more choices or adjustable parameters in the shape of our mechanical primitives, so as to better match the appearance of the provided model. P2 and P6 thought that the system takes the pivot point as the center position to place the mechanism, which leads to confusion. As introduced before, the pivot point is used as the center point of the bottommost surface of the generated mechanism. P4 and P8 thought the system supports importing two models and combining them with different mechanical parts respectively. Since RoamFab can only import one model at present, they suggested adding this feature because sometimes the models are originally separate. P5 suggested the ability to split the model along the shape of the curved surface, so that the original shape is not modified. P1, P4, and P5 would like to have animations to visualize how the mechanical movement is achieved, which helps them verify that the reconfiguration of mechanisms is functioning as expected. All these suggestions fertilize opportunities for future work.

9. Discussions, limitations & future work

From the user study, we observe two major obstacles that limit the use of RoamFab by novice users. The first is that users lack the skills to manipulate Rhino's interface, such as specifying a point. This can be solved by generating some points and letting the user choose. For example, the system can recommend the vertices resulting from the intersection of a surface and the target model, and the surface can be generated by computing the convex hull of the target model. Besides, the precise selection of points may not be necessary in the future. The system can automatically snap a point to the model by calculating the Euclidean distance after uniformly sampling its edges. We also found that displaying more views helps users understand the spatial relationship. The user can then drag the point in the viewport.

Next, even though our participants already understood how the mechanism works, they generally found it challenging to determine where and how to configure it, especially when the mechanism needs to work with both parts of the original model. Although they can keep trial and error to

find reasonable results, the process can be frustrating. As one participant mentioned that she needs more learning examples to draw inferences. Therefore, more prior knowledge about the context in using mechanisms should be integrated into RoamFab. It is possible to provide more specific guidance on how to configure a mechanism. For example, by highlighting colors, the system can interactively prompt users which part of the model would be combined with the ratchet wheel or the inner pawl. Future work should also detect the collision between different parts of the model based on the degree of freedom of mechanical motion. In addition, providing animations to simulate the mechanical movements is also beneficial for users to validate their results.

In the future, the shape of mechanical primitives could be customized to make them more compatible with user-provided models. More mechanisms could also be added to RoamFab, such as the cam and follower mechanism that converts rotary motion into linear motion, and the Geneva mechanism for producing intermittent rotary motion. Our computational pipeline makes it flexible to reconfigure new parameterized mechanical primitives to given 3D models, which may need different geometric processing techniques to satisfy their mechanical constraints, though. For the structural optimization section, we only considered the bending strength of a specific mechanical part. Further work should consider using topology optimization (Liu et al., 2018) to provide an interactive analysis of the structural strength of the whole object while taking into account the print orientation at the same time. Currently, we use Ultimaker Cura to slice the object with different infill blocks. This process can be automated by generating G-code directly in Grasshopper. Besides, our mechanical experiment focuses on the mechanical properties of PLA due to its easy printability without a significant smell. Since PETG and ABS have become increasingly popular for their strength and durability, we should consider incorporating these materials within RoamFab.

As described by Ledo et al. (2018), ideal toolkits should enable creative exploration of novel design spaces. We also see the potential of RoamFab in inspiring end-users to author personalized artifacts that can assist their daily-living tasks or be used for recreation. Our participants appreciated the easy access to configure mechanisms, which helps them pick up some interesting ideas that they “once found troublesome and gave up.” RoamFab also enables end-users to expand new solutions. Two of our participants felt a sense of accomplishment because they proposed two solutions. Meanwhile, we assume that the fabricated results produced by the user will further inspire each other in makers communities. They may find it is easy to create interactive artifacts instead of being limited to downloading and printing static models from online communities.

10. Conclusion

This study presents RoamFab, a design tool that empowers novices makers and designers to rapidly convert existing 3D shapes into fabrication-ready models with desired mechanical movements. We first parameterized four mechanical

primitives frequently used for 3D printing objects, which offloads the complexity in designing mechanisms and saves efforts for testing their 3D printing tolerances. Based on a mechanical experiment, we present an optimization procedure to guarantee that an object can provide enough bending strength before 3D printing. Finally, we introduced a computational pipeline that reconfigures our mechanical primitives into arbitrary 3D models in user-customized ways. The ability to fit the generated geometry to the dimensions of the target model further simplifies the authoring process for users. From the user study, we have seen that participants could create a wide range of functional objects with our design workflow. We hope RoamFab can facilitate the creative exploration of end-users in designing their personalized artifacts.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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